

Joint High-Speed Communications, Safety Power Delivery and Temperature Sensing over Multicore Fibers

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Abstract We report the first measurement of temperature variations induced by PoF using FBGs. Tests on a 7-core MCF with 25.67 Gbps 5G-NR ARoF and 25 Gbps NRZ transmission, delivering 360 mW at 2 km, show that 2 W PoF increases core temperature by 16 ± 1 °C. ©2026 The Author(s)

Introduction

The rapid evolution of mobile communications toward 5G and future 6G architecture demands solutions that enhance network control and improve system efficiency. In this context, Centralized Radio Access Network (C-RAN) architectures have emerged as a key approach, enabling centralized processing and the deployment of distributed Remote Units (RUs) [1, 2]. Analog Radio over Fiber (ARoF) plays a key role by enabling the direct transport of radiofrequency signals in analog form, thereby eliminating digitalization processes and complex digital processing at the RUs. This significantly reduces RU complexity and power consumption while supporting wideband signals [3].

Recent advances in Power over Fiber (PoF) techniques, together with increasing Photovoltaic Power Converters (PPCs) efficiency, have made PoF to RUs practically feasible. Furthermore, Multicore Fibers (MCFs) offer additional advantages by increasing transmission capacity and enabling the physical separation of data and PoF, thereby mitigating signal degradation and nonlinear effects.

Power monitoring techniques in MCF links are essential to ensure safe and reliable operation. Several approaches have been reported, including backscattering detection [4], crosstalk-based monitoring [5], and the use of Tilted Fiber Bragg Gratings (TFBGs) near the RU [6].

Joint communication and sensing is attracting increasing attention, including the analysis of potential optical transmission degradation [7]. Temperature monitoring is critical, among other factors, for accurate power delivery assessment and safety in PoF systems. FBG temperature sensing with low PoF signals (tenths of mW) is reported in [8]; however, only ambient temperature variations were measured in a 20 km single-mode fiber link with 1.25 Gbps data transmission. The use of TFBGs inscribed in MCFs to measure temperature variations while monitoring the PoF signal was proposed in [6], but it was not experimentally validated.

In this work, we experimentally evaluate a joint sensing, power delivery and high throughput radio

and fixed data transmission link. Measuring for the first time the temperature rise in a MCF, when delivering up to 2 W, with a FBG and a TFBG centered at 1538 nm and 1390 nm respectively. Ambient temperature variations are also measured together with ARoF and NRZ data signals in both cases. In addition, the optical power delivered to the RU is passively monitored by measuring the backcrosstalk from TFBG at 1580 nm.

Experimental Setup

Fig. 1 shows the experimental setup implemented over a 2 km, 7-core MCF, with two TFBGs and a FBG inscribed at the end of the fiber, close to the RUs. A Signal Quality Analyzer (SQA) generates NRZ signals at 12.5 Gbps with an SFP transceiver at 1550 nm. 5G NR signals are generated using two Vector Signal Generators (VSGs) with configuration parameters summarized in the table shown in Fig. 1. VSG1 is configured in band n258 at a carrier frequency of 26 GHz and modulates the optical transmitter of the Lightwave Component Analyzer (LCA) centered at 1550 nm. The resulting optical signal is coupled to cores 5 and 6 with a 1×2 splitter. At the receiver, the signal is detected by the LCA, and amplified in RF before being processed by a Vector Network Analyzer (PNA).

VSG2 operates in FR3 at 13 GHz. Four laser diodes (LD) centered at 1310, 1530, 1550, and 1570 nm are externally modulated to evaluate transmission performance across different spectral bands. Each optical carrier passes through a polarization controller and is modulated by a dedicated Mach-Zehnder Modulator (MZM).

The 5G NR signal at 1310 nm and the NRZ signals are launched into cores 1 and 7 of the MCF using a wavelength division multiplexer (WDM) and a 1×2 splitter. In the case of the signals launched into core 7, an optical circulator is also inserted to collect the reflected signals for the monitoring and sensing process, based on detecting the energy reflected by the TFBG inscribed in the O- and L-bands.

The remaining 5G NR signals are injected into cores 2 and 3. At the remote end, 5G NR signals are detected using a photodetector (PD3), followed

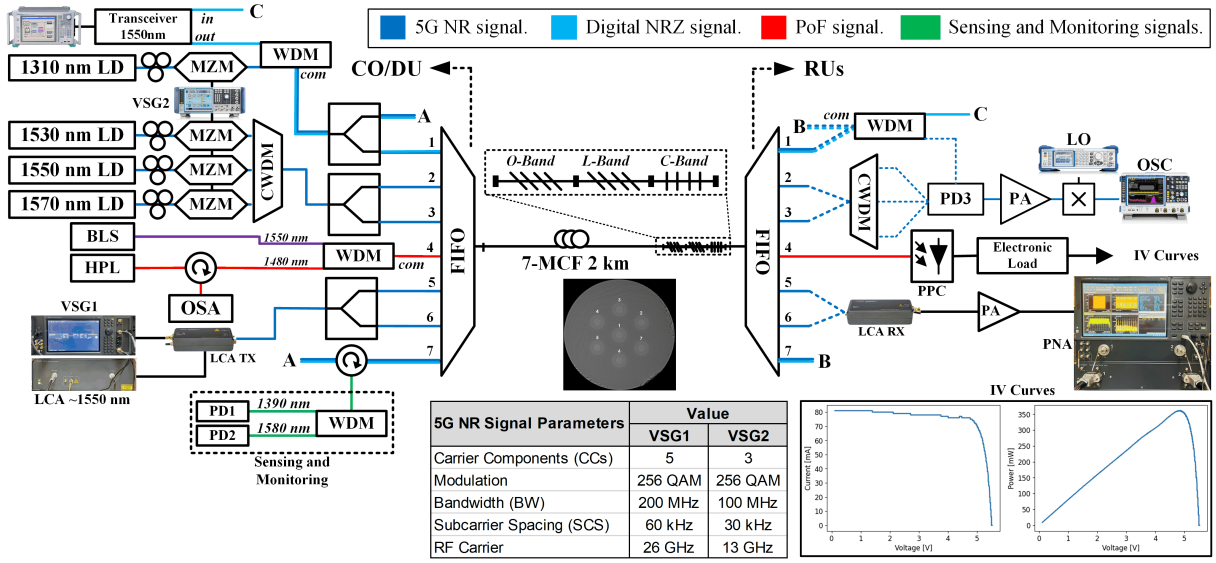


Fig. 1: Experimental setup. Note that points A–A, B–B, and C–C correspond to the same location. The dashed lines represent connections that can be switched to measure different data transmission per core.

by an RF amplifier. A mixer and a local oscillator are then used to downconvert the signals to fit within the 2 GHz oscilloscope bandwidth.

A high-power laser (HPL) generates the PoF signal with up to 2 W of optical power. An optical circulator protects the HPL from back-reflections and enables spectral analysis of the FBG inscribed in the core delivering PoF. A broadband light source (BLS) is used to interrogate the FBG, and an optical spectrum analyzer (OSA) measures the reflected spectrum. Optical-to-electrical conversion is performed using photovoltaic power converters (PPCs). The I–V curves in Fig. 1 show the maximum optical power and the corresponding generated electrical power, with the PPC achieving up to 360 mW of electrical output.

Experimental results

First, we measure the spectral shift of the FBG and TFBG (1390 nm) inscribed in the MCF, in order to evaluate their thermal response and temperature measurement capacity when the HPL is switched off. A hot plate is placed at a known temperature, and a controlled temperature variation of 20 °C is applied from 40 °C as the starting point. The plate is positioned over the FBG or TFBG, enabling heat transfer that induces a shift in the Bragg wavelength. Temperature control system integrates an infrared camera (FLIR AX5) with 1 °C resolution. The calibration results are shown in Fig.2 and the sensitivity is derived.

Subsequently, an HPL power sweep is performed from 23 to 33 dBm without the hot plate. In both cases, a spectral shift is observed, which is associated with a temperature increase of around 16–17 °C based on previously determined sensitivities. These results corroborate that PoF delivery induces a temperature increase and that the TFBG can be used for either ambient or PoF-induced

thermal analysis without requiring an FBG, thereby avoiding the need for a circulator.

The temperature sensing uses the 1390 nm TFBG, as it provides the highest backcrosstalk power since it is inscribed at the second harmonic of the HPL [6]. Meanwhile the TFBG at 1580 nm is used for monitoring as its backcrosstalk power versus the set HPL power exhibits a highly stable behavior, as shown in Fig. 3. To validate the monitoring technique under operation, simultaneous PoF delivery, NRZ digital signals compatible with PON, and 5G NR ARoF signals are transmitted. Concurrently, HPL power sweeps are performed in the range from 23 to 33.5 dBm with 0.1 dB step, enabling the evaluation of a large number of power operating points. The monitoring signal is measured at the input of core 7 using PD2 and a data acquisition card with a sampling rate of 2 kSamples/s over a duration of 1 minute.

Results of successful monitoring of PoF delivery are shown in Fig. 3. The voltage is referenced to the laser-off state ($\Delta V = 0$). The measured voltage variations versus HPL power sweep has a quadratic dependence with $R^2 = 0.99$. At low HPL power levels, larger uncertainties are observed because the power reflected by the TFBG becomes comparable to the other propagating carriers.

To evaluate the data quality and enable efficient transmission, data delivery is implemented across all cores of the MCF, except for core 4, which is dedicated to PoF delivery. This approach avoids introducing additional elements into this link and thus prevents any degradation of system efficiency, reaching a total throughput of 50.67 Gbps (25 Gbps NRZ + 25.67 Gbps 5G NR). The transmitted 5G NR signals are evaluated using the Error Vector Magnitude (EVM). EVM measurements of 5G NR signals while varying the HPL from 23 dBm

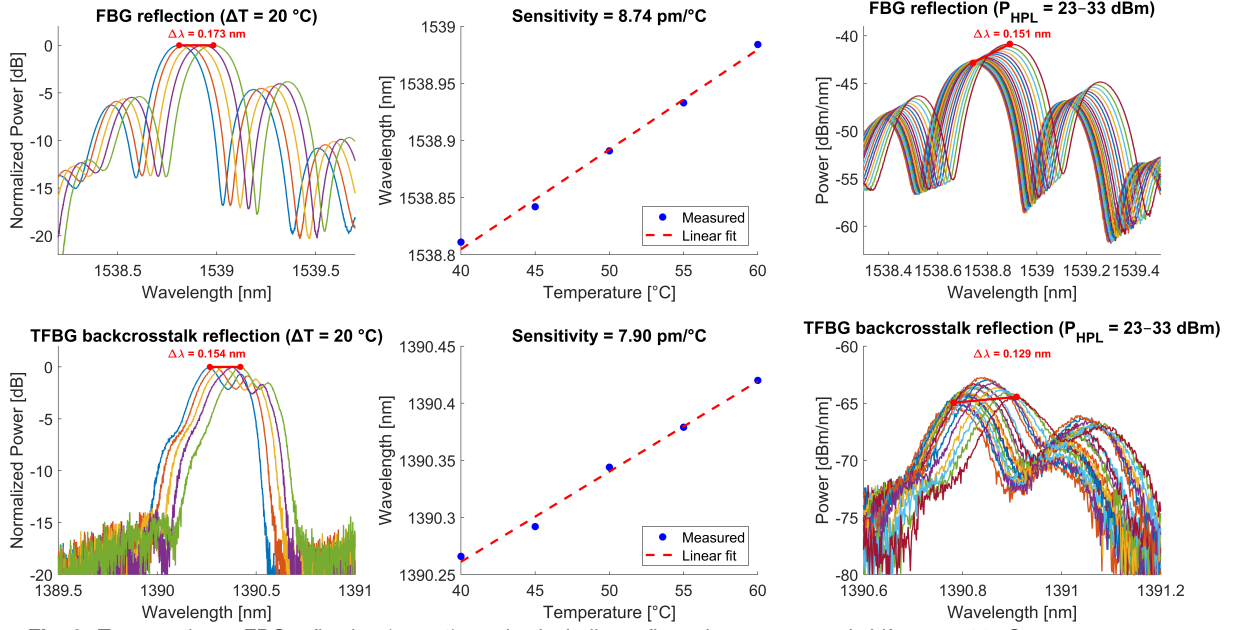


Fig. 2: Top row shows FBG reflection (core 4) results, including reflected power spectral shift over a 20 °C temperature range induced by external heating, sensitivity analysis, and spectral shift induced by HPL power delivery without external heating. Bottom row presents the corresponding TFBG backcrossstalk reflection (core 7) measurements.

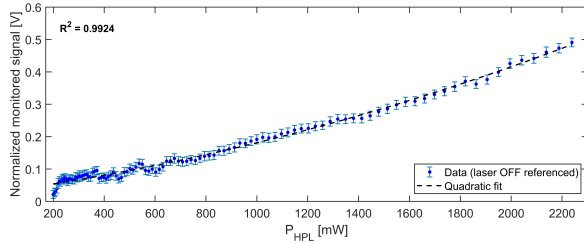


Fig. 3: Monitoring of the backcrossstalk power from the 1580 nm TFBG together with the fitted curve as a function of P_{HPL} .

to 33 dBm with 1 dB step, for each optical carrier and their corresponding subcarriers are shown in Fig. 4 a-e. The RF spectrum of the single ARoF signal with 5 CC providing 1 GHz bandwidth is shown in Fig. 4 f. Error-free communication ($\text{BER} < 10^{-12}$) of NRZ signal over the entire power range are also obtained. Results show that the presence of PoF signals does not significantly affect the quality of either the 5G NR transmissions (remaining within the EVM limits defined by the 3GPP standard) or the NRZ transmissions while achieving record throughputs of 50.67 Gbps in monitored PoF, ARoF and NRZ data links. [9].

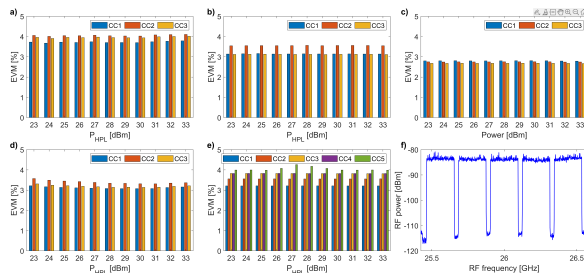


Fig. 4: EVM of 5G NR ARoF signals component carriers at different cores vs swept P_{HPL} . (a-c) at 1530, 1550, and 1570 nm in core 2, (d) at 1310 nm in core 1, (e) at 1550 nm in core 5. (f) Electrical spectrum of ARoF signal in core 5.

Conclusions

In this work, a passive sensing and monitoring technique for temperature and delivered power at a remote node are demonstrated along with high speed communications. The temperature variations related to energy delivery by PoF are measured for the first time. The approach is validated through the simultaneous transmission of PoF, PON-compatible NRZ digital signals, and 5G NR ARoF signals over a 2 km 7-core MCF. An aggregated throughput of 50.67 Gbps is successfully achieved, considering data rates of 25 Gbps for NRZ signals and 25.67 Gbps for 5G NR ARoF signals, while delivering up to 360 mW electrical power at the RUs. Experimental results show that the optical power propagating through the fiber induces a temperature increase of $16 \pm 1\text{ }^{\circ}\text{C}$, which is consistently measured using both a reference FBG and the proposed TFBG sensor. Furthermore, the proposed monitoring technique exhibits a quadratic response of the measured signal with respect to the delivered power, enabling accurate estimation. This monitoring and sensing capability is achieved without degrading the performance of the data transmission, which remains compliant with the relevant communication standards. The inscribed TFBG are used for PoF monitoring and simultaneous temperature sensing while providing more than 50.6 Gbps mobile and fixed data communications and can be extend to simultaneous measurement of other parameters.

Acknowledgements

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References

- [1] N. Chen and M. Okada, "Toward 6G Internet of Things and the Convergence With RoF System", *IEEE Internet of Things Journal*, pp. 8719–8733, 2021.
- [2] C. Ranaweera, J. Kua, I. Dias, E. Wong, C. Lim, and A. Nirmalathas, "4g to 6g: Disruptions and drivers for optical access [invited]", *Journal of Optical Communications and Networking*, vol. 14, no. 4, A143–A153, 2022. DOI: 10.1364/JOCN.451632.
- [3] S. Rommel et al., "Towards a Scaleable 5G Fronthaul: Analog Radio-over-Fiber and Space Division Multiplexing", *Journal of Lightwave Technology*, pp. 5412–5422, 2020.
- [4] J. Barco-Álvarez, R. Altuna, and C. Vázquez, "Backscattering of Crosstalk for Monitoring Power over Fiber Co-transmission with 5G NR Analog Radio Over Fiber and NRZ Signals over Multicore Fiber", in *ECOC 2025*, Copenhagen, Denmark, Sep. 2025.
- [5] R. Altuna, J. D. López-Cardona, and C. Vázquez, "Monitoring of Power Over Fiber Signals Using Intercore Crosstalk in ARoF 5G NR Transmission", *Journal of Lightwave Technology*, vol. 41, no. 23, pp. 7155–7161, Dec. 2023. DOI: 10.1109/JLT.2023.3300184.
- [6] J. Barco-Álvarez, R. Altuna, D. S. Montero, D. Barrera, S. Sales, and C. Vázquez, "Safety Power and 5G NR Analog Radio Over Multicore Fiber Transmission With Tilted Fiber Bragg Gratings", *Journal of Lightwave Technology*, 2025. DOI: 10.1109/JLT.2025.3589657.
- [7] L. Rapp and D. Kroushkov, "Transmission Performance Degrading Effects in Joint Communication and Fiber Sensing Scenarios", in *2025 IEEE Photonics Society Summer Topicals Meeting Series (SUM)*, Berlin, Germany, 2025. DOI: 10.1109/SUM65312.2025.11121749.
- [8] X. Gu et al., "Passive-active hybrid fbg-power-over-fiber system for integrated sensing, power delivery, and communication", *Opt. Express*, vol. 33, no. 20, pp. 42 343–42 357, Oct. 2025. DOI: 10.1364/OE.568129.
- [9] R. Altuna, J. Barco-Álvarez, and C. Vázquez, "Power Over Fiber Co-Transmission With 39.5 Gpbs Mixing 5G NR Analog Radio Over Fiber and NRZ Signals Over Multicore Fiber", *Journal of Lightwave Technology*, pp. 1–7, 2025. DOI: 10.1109/JLT.2025.3534921.